

Technical Paper: Universal Anomaly Detection via Deep Hybrid-Cloud Autoencoders

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1. Abstract

Anomaly detection is a critical component of modern automated systems, yet traditional supervised methods often fail to capture zero-day threats or rare pathological deviations due to a lack of labeled training data. This paper proposes a **Universal Anomaly Engine**—a multi-domain platform leveraging unsupervised **Deep Autoencoders**. By training strictly on "normal" data distributions, the system learns to reconstruct expected patterns. Deviations are quantified via Reconstruction Error (MSE), providing a robust, domain-agnostic detection mechanism. We demonstrate successful deployment across financial, medical, and network security domains using a high-performance **Hybrid-Cloud Architecture**.

2. Introduction

As data complexity increases, the "Needle in the Haystack" problem becomes more pronounced. Our research focuses on **Unsupervised Representation Learning**. By using Autoencoders (AE), we compress data into a low-dimensional "Latent Space" and then attempt to reconstruct the original input. This process naturally filters out noise and forces the model to learn the fundamental structure of the data.

3. Methodology & System Architecture

3.1 Neural Architectures

We utilized three distinct PyTorch-based architectures tailored to domain-specific data types:

- **Dense Dimensionality Reduction (Finance):** A 5-layer fully connected AE designed to process 30-feature PCA transaction vectors.
- **Spatial Convolutional AE (Medical):** A Conv2D architecture with pooling/unpooling layers to analyze 64x64 Grayscale Chest X-Rays.
- **Deep Feature Extractor (Network):** A high-depth bottleneck AE (77 to 8 dimensions) to map complex packet metadata.

3.2 Hybrid-Cloud Deployment

To achieve both high-fidelity UI and low-latency inference, we implemented a **Split Architecture**:

1. **Inference Layer (Hugging Face):** Optimized PyTorch weights hosted on Python 3.10 Gradio instances.
2. **Display Layer (Vercel):** A sophisticated React dashboard providing an "Enterprise Elegant" dashboard for end-users.

4. Empirical Results & Discussion

Our testing focused on the **Seperation Margin** between normal and anomalous instances:

Domain	Baseline (Normal) MSE	Anomaly (Pathology) MSE	Detection Confidence
Banking	< 0.21	> 0.69	High (Deterministic)
Medical	< 0.0036	> 0.0480	Very High (Visual Signal)
Network	< 0.15	> 0.62	High (Signature Match)

Key Finding: The Medical Diagnostic model showed the strongest "Signal-to-Noise" ratio, with pathological scans exhibiting over **13x** the reconstruction error of healthy lungs.

5. Conclusion & Future Work

The Universal Anomaly Engine proves that unsupervised deep learning can provide a unified defense across disparate domains. Future research will focus on **Real-Time Latent Space Visualization** and **Federated Learning** to improve model privacy in healthcare environments.

6. References

1. Goodfellow, I., et al. (2016). Deep Learning. MIT Press.
2. Kaggle Datasets: Credit Card Fraud, Pediatric Pneumonia, & NSL-KDD.
3. Pytorch Documentation: Unsupervised Learning with Autoencoders.